

A Technical Introduction to Perceptron

$$y = \text{sign}(w \cdot x + b)$$

$$w \leftarrow w + \eta(y - \hat{y})x$$

Section 1

A single perceptron is a linear classifier. It can only reach a stable state if all input vectors are classified correctly. In case the training set D is not linearly separable, i.e. if the positive examples cannot be separated from the negative examples by a hyperplane, then the algorithm would not converge since there is no solution. Hence, if linear separability of the training set is not known a priori, one of the training variants below should be used. Detailed analysis and extensions to the convergence theorem are in Chapter 11 of Perceptrons (1969). Linear separability is testable in time $\min(O(n d / 2), O(d^2 n), O(n d - 1 \ln n))$, where n is the number of data points, and d is the dimension of each point. If the training set is linearly separable, then the perceptron is guaranteed to converge after making finitely many mistakes. The theorem is proved by Rosenblatt et al.

Section 2

Learning again iterates over the examples, predicting an output for each, leaving the weights unchanged when the predicted output matches the target, and changing them when it does not. The update becomes:

Section 3

Further reading Aizerman, M. A. and Braverman, E. M. and Lev I. Rozonoer. Theoretical foundations of the potential function method in pattern recognition learning. Automation and Remote Control, 25:821–837, 1964. Rosenblatt, Frank (1958), The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain, Cornell Aeronautical Laboratory, Psychological Review, v65, No. 6, pp. 386–408. doi:10.1037/h0042519. Rosenblatt, Frank (1962), Principles of Neurodynamics. Washington, DC: Spartan Books. Minsky, M. L. and Papert, S. A. 1969. Perceptrons. Cambridge, MA: MIT Press. Gallant, S. I. (1990). Perceptron-based learning algorithms. IEEE Transactions on Neural Networks, vol. 1, no. 2, pp. 179–191. Olazaran Rodriguez, Jose Miguel. A historical sociology of neural network research. PhD Dissertation. University of Edinburgh, 1991. Mohri, Mehryar and

Rostamizadeh, Afshin (2013). Perceptron Mistake Bounds arXiv:1305.0208, 2013. Novikoff, A. B. (1962). On convergence proofs on perceptrons. Symposium on the Mathematical Theory of Automata, 12, 615–622. Polytechnic Institute of Brooklyn. Widrow, B., Lehr, M.A., "30 years of Adaptive Neural Networks: Perceptron, Madaline, and Backpropagation," Proc. IEEE, vol 78, no 9, pp. 1415–1442, (1990). Collins, M. 2002. Discriminative training methods for hidden Markov models: Theory and experiments with the perceptron algorithm in Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP '02). Yin, Hongfeng (1996), Perceptron-Based Algorithms and Analysis, Spectrum Library, Concordia University, Canada

Section 4

"cross-coupling" (connections between units within the same layer) with possibly closed loops, "back-coupling" (connections from units in a later layer to units in a previous layer), four-layer perceptrons where the last two layers have adjustable weights (and thus a proper multilayer perceptron), incorporating time-delays to perceptron units, to allow for processing sequential data, analyzing audio (instead of images). The machine was shipped from Cornell to Smithsonian in 1967, under a government transfer administered by the Office of Naval Research.

Section 5

Multiclass perceptron Like most other techniques for training linear classifiers, the perceptron generalizes naturally to multiclass classification. Here, the input x and the output y are drawn from arbitrary sets. A feature representation function $f(x, y)$ maps each possible input/output pair to a finite-dimensional real-valued feature vector. As before, the feature vector is multiplied by a weight vector w , but now the resulting score is used to choose among many possible outputs:

Section 6

A single perceptron can learn to classify any half-space. It cannot solve any linearly nonseparable vectors, such as the Boolean exclusive-or problem (the famous "XOR problem"). A perceptron network with one hidden layer can learn to classify any compact subset arbitrarily closely. Similarly, it can also approximate any compactly-supported continuous function arbitrarily closely. This is essentially a special case of the theorems by George Cybenko and Kurt Hornik.

Section 7

Boolean function When operating on only binary inputs, a perceptron is called a linearly separable Boolean function, or threshold Boolean function. The sequence of numbers of threshold Boolean functions on n inputs is OEIS A000609.

The value is only known exactly up to $n = 9$ case, but the order of magnitude is known quite exactly: it has upper bound $2^{n^2 - n \log_2 n} + O(n)$ and lower bound $2^{n^2 - n \log_2 n} - O(n)$. Any Boolean linear threshold function can be implemented with only integer weights. Furthermore, the number of bits necessary and sufficient for representing a single integer weight parameter is $\Theta(n \ln n)$.

Section 8

The following simple proof is due to Novikoff (1962). The idea of the proof is that the weight vector is always adjusted by a bounded amount in a direction with which it has a negative dot product, and thus can be bounded above by $O(\sqrt{t})$, where t is the number of changes to the weight vector. However, it can also be bounded below by $O(t)$ because if there exists an (unknown) satisfactory weight vector, then every change makes progress in this (unknown) direction by a positive amount that depends only on the input vector.

Section 9

Although the perceptron initially seemed promising, it was quickly proved that perceptrons could not be trained to recognise many classes of patterns. This caused the field of neural network research to stagnate for many years, before it was recognised that a feedforward neural network with two or more layers (also called a multilayer perceptron) had greater processing power than perceptrons with one layer (also called a single-layer perceptron). Single-layer perceptrons are only capable of learning linearly separable patterns. For a classification task with some step activation function, a single node will have a single line dividing the data points forming the patterns. More nodes can create more dividing lines, but those lines must somehow be combined to form more complex classifications. A second layer of perceptrons, or even linear nodes, are sufficient to solve many otherwise non-separable problems. In 1969, a famous book entitled *Perceptrons* by Marvin Minsky and Seymour Papert showed that it was impossible for these classes of network to learn an XOR function. It is often incorrectly believed that they also conjectured that a similar result would hold for a multi-layer perceptron network. However, this is not true, as both Minsky and Papert already knew that multi-layer perceptrons were capable of producing an XOR function. (See the page on *Perceptrons* (book) for more information.) Nevertheless, the often-miscited Minsky and Papert text caused a significant decline in interest and funding of neural network research. It took ten more years until neural network research experienced a resurgence in the 1980s. This text was reprinted in 1987 as "Perceptrons - Expanded Edition" where some errors in the original text are shown and corrected.